

Tutorial 10
Convolutional Neural Networks and Recurrent Neural Networks

Summary of Key Concepts

In this tutorial, we will discuss and explore the following learning points from Lecture:

1. CNN
 - (a) Convolution kernel
 - (b) CNN receptive field
 - (c) Padding and stride
 - (d) Design choice
2. RNN Design choice
3. CNN vs RNN

A ConvNets

1. You are given an image $\mathbf{x} \in \mathbb{R}^{5 \times 5}$ and a filter $\mathbf{W} \in \mathbb{R}^{3 \times 3}$. **No** extra padding is added.

0.1	0.2	0.1	0.1	0
0.8	0.9	1	1	0.9
1	1	1	1	1
0.9	1	1	0.8	1
0	0.1	0.1	0.2	0

1	1	1
0	0	0
-1	-1	-1

Figure 1: (left) The image $\mathbf{x}$. (right) The filter $\mathbf{W}$.

Get the output feature map from this convolution and write down its output dimensions if the kernel moved with a stride of 1×1 . No Padding or Pooling op-

eration required. Additionally, what do you think is the purpose of this filter?

- Now, let's say you have access to a very deep, large CNN model. We feed a single image to the network. Each image has $3(C)$ channels (RGB) with a height and width of 224×224 ($H = 224, W = 224$). Here our input is a 3-dimensional tensor ($H \times W \times C$). The first layer of the big CNN is a convolutional Layer with 96 (C_1) kernels which has a height and width of 11, and each kernel has the same number of channels as the input channel. The stride is 4×4 and no padding is used. Calculate the output size $H_1 \times W_1 \times C_1$ after the first convolutional Layer.
- In most of Deep Learning libraries such as PyTorch, images are fed together as a batch B . B can take values such as 8, 16, 32, 64. Comment on the output shape if we feed the large CNN in part (b) with a batch of images. What are the advantages of using a batch of images rather than a single image?

B CNN Receptive field

The receptive field refers to the specific region of the input image that a particular neuron in the network is influenced by. Essentially, it's the area of the input that contributes to the output of a neuron in a convolutional layer. It gives us an idea of where we're getting our results from as data flows through the layers of the network. To further illuminate the concept let's have a look at this illustration

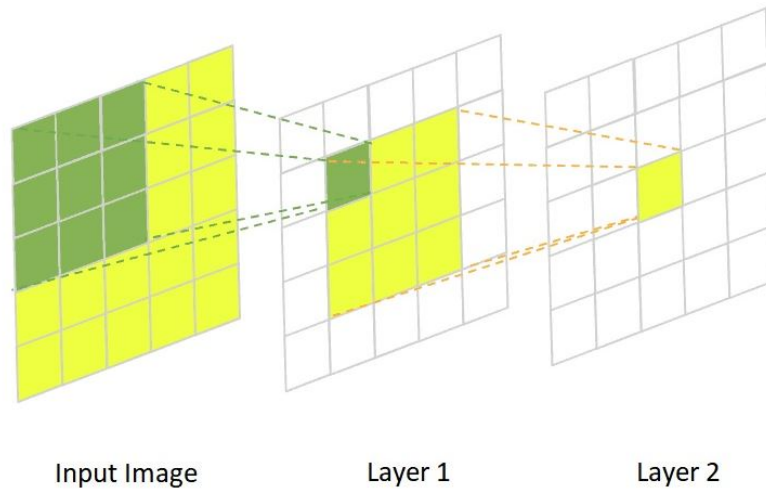


Figure 2: Two-layered CNN with a 3x3 kernel

The green area in the input image represents the receptive field (3x3) of one neuron in Layer 1, as the output of this neuron is influenced by a 3x3 pixel region in the input image. The yellow area in the input image marks the receptive field of one neuron in Layer 2, where a 5x5 pixel area from the input image is considered.

- Consider a 2-layer convolutional neural network (CNN) with the following configuration:
 - An initial input matrix with size 9×9

- Layer 1: Uses a kernel of size 5×5 and a stride of 2×2 .
- Layer 2: Uses a kernel of size 2×2 and a stride of 1×1 .

Calculate the receptive field sizes of neurons in the first and second layers.

2. How does an increase in the receptive field affect the performance of a Convolutional Neural Network (CNN)?

C RNN Design

1. What type of RNN model does Image captioning use? What characteristics make this a standard model for Image captioning? Can you think of any other examples for this type of RNN?
2. What type of RNN model does stock market prediction use? What characteristics make this a good model for stock prediction? Can you think of any other examples for this type of RNN?
3. What type of RNN model does language translation use? What characteristics make this a good model for language translation? Can you think of any other examples for this type of RNN?

D CNN vs RNN

1. Let's apply what you learnt to a specific example, performing sentiment analysis on Covid-19 posts on twitter. Explain what characteristics of RNN make it a good model for sentiment analysis and which RNN model you want to use to tackle this problem.
2. Given the same context in part (1), would it be possible to perform sentiment analysis using CNN? Explain why or why not.
3. In part (2), we've looked at ConvNets for sequences (sentiment analysis of text, for instance). Now, let's examine RNNs for image processing.

"Tokens" are what constitute a sequence; words are tokens in sentences, for instance. Usually, when it comes to words, we represent them as vectors. We also know RNNs take in sequences (temporally-related collection of tokens). If we had to treat an image as a sequence, what would you do to allow RNNs to classify images? In other words, how can you tokenize an image? Your final method of "tokenizing" an image should ensure an RNN can be fed these tokens as inputs.

Express your creativity in coming up with different ways to go about doing so.